An Auditable, Leakage-Hardened Machine-Learning System for Constraint-Bound Marine Coating Design

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***Abstract*—Machine learning for materials discovery routinely fails its consumers in two ways: benchmark results that do not survive leakage audits, and formulation recommendations that cannot be traced, reproduced, or tested by a laboratory. We present a two-stage system for constraint-bound coating design that treats *auditability as the primary design constraint*. Stage one is a multi-property prediction engine evaluated under a referee-hardened protocol: SHA256-committed data splits and feature whitelists, paired cluster-block bootstrap inference with pre-registered minimum effect sizes, label-permutation (y-scramble) leakage canaries, false-discovery-rate control, and conformal prediction intervals with measured coverage. Stage two is a deterministic formulation compiler that converts a customer technical envelope into laboratory-executable candidate formulations with per-claim evidence tiers (REAL / MODELED / ABSENT) and fail-closed gates. On the 7,367-polymer PolyMetriX glass-transition benchmark our commercially-clean stacked ensemble is evaluated against tuned gradient-boosting, graph-neural, and tabular-foundation-model baselines (TabPFN-2/2.5, TabFM) under identical frozen splits, including cluster-grouped splits that repair a defective scaffold protocol we document in prior work. We report wins and non-wins symmetrically, quantify a physics-grounded rediscovery signal for known high-performance epoxy hardeners against property-matched decoys (AUROC 0.7252, permutation p =0.008549572521373932), and measure single-core screening throughput of 7,672 candidates/s. We argue that the resulting evidence discipline, not any single leaderboard number, is the load-bearing contribution for safety-critical formulation programs.**

I. Introduction

Naval coating programs specify products through binding technical envelopes: fixed chemistry class, mix ratio, volume solids, volatile-organic-compound (VOC) ceilings, film-build windows, pot life, and shelf life. A machine-learning system that proposes formulations for such a program must satisfy two masters. It must search an enormous chemical space quickly, and every number it emits must survive hostile review — by domain experts, by procurement, and ultimately by a wet laboratory whose measurements do not negotiate.

This paper reports a system built under an unusual governance regime: eleven internal red-team rounds, each followed by verified corrections, with the standing rule that *no claim may exceed its evidence*. Three consequences distinguish the work from typical ML-for-materials reports. First, all validation is performed under frozen, hash-committed splits with an explicit feature whitelist, because our own audit found the benchmark's metadata columns contain the target itself (Section IV). Second, model comparisons use paired cluster-block bootstrap inference with pre-registered minimum effect sizes above the label quantization floor, and losses are reported next to wins. Third, the formulation stage never presents a modeled quantity as measured: every claim carries an evidence tier, and unproven properties fail closed to a laboratory test plan rather than passing silently.

II. Related Work

Constrained mixture design descends from Scheffé's simplex treatment and Cornell's canonical text. Closed-loop Bayesian optimization has reached formulation chemistry, and self-driving laboratories increasingly close the synthesis loop. Polymer informatics provides property predictors (polyBERT, RadonPy, chemprop's directed message-passing networks), and tabular foundation models (TabPFN family; TabFM) now offer strong zero-shot baselines for small-to-medium tabular problems. Our contribution is not a new learner: it is the auditable *system* — envelope compilation with feasibility-by-construction on the mix-ratio/solids surface, evidence-tiered outputs, and a validation protocol hardened against the leakage and baseline-handicap failures that an adversarial review of our own prior work uncovered.

III. System Overview

A. Stage 1: Multi-property prediction engine

The engine predicts coating-relevant properties from polymer or molecule structure using feature-block ensembles: curated polymer-domain descriptors, general cheminformatics descriptors, hashed circular fingerprints, and pretrained polymer embeddings, combined by out-of-fold stacking into a ridge meta-learner. Two configurations are maintained deliberately: a *commercially-clean stack* (Apache/BSD/MIT components only) that constitutes the product path, and a *ceiling* configuration that adds non-commercially-licensed foundation-model predictions as meta-features to quantify headroom. Licensing is treated as a first-class experimental variable because several state-of-the-art tabular foundation models restrict commercial use.

B. Stage 2: Formulation compiler

A deterministic compiler parses a customer technical envelope into a machine-readable constraint set, solves solvent splits in closed form so that every generated candidate lies

exactly on the mix-ratio and volume-solids surface, enforces the remaining envelope gates fail-closed, and ranks survivors with published weights. Outputs are laboratory-executable blueprints with a qualification/acceptance matrix in which each test carries a method and a pass/fail criterion sourced either from the envelope itself or from the public military specification — criteria that are not publicly resolvable are marked for confirmation against the controlled document rather than guessed. Recommendation robustness is stress-tested by weight perturbation; the system reports a co-lead set with the tie-break explicitly deferred to laboratory data when the margin lies below model uncertainty. Formulation identities and ratios are not disclosed in this external document.

IV. Experimental Protocol

Data. Polymer glass transition: PolyMetriX curated set (7,367 polymers). Multi-property: RadonPy PI1070 (1,077 polymers; molecular-dynamics-computed targets — a surrogate-of-simulation setting, with each target's replicate noise floor reported). Thermoset transfer: 15 literature-verified cured epoxy-network systems. All datasets are public.

Leakage controls. The PolyMetriX distribution includes metadata columns that literally contain the target (e.g. a tg-range field equal to the label). All models receive features exclusively through a SHA256-committed whitelist of 100 structure-derived descriptors plus independently computed feature blocks; label-permutation canaries confirm that scrambled targets collapse to trivial error on every split family.

Splits. Five seeded random splits; grouped leave-polymer-class-out (13 folds); and Butina sphere-exclusion clustering on capped repeat units (2,622 clusters; largest cluster 2% of data; no test fold contains a cluster exceeding 10% of its rows). We document that a conventional Bemis–Murcko scaffold split is defective for polymer repeat units: 13.6% of the corpus collapses into a single empty-scaffold bucket, which a largest-first assignment then places entirely in test. Prior results built on such splits overstate generalization error structure.

Inference. Paired cluster-block bootstrap (4,000 resamples at the group level) on per-row absolute error differences; group-level sign-permutation tests; Benjamini–Hochberg false-discovery-rate control across each comparison family; a pre-registered minimum meaningful effect of 1.0 K (85% of glass-transition labels are integer-degree quantized); split-conformal 90% prediction intervals with measured coverage.

Baselines. Tuned LightGBM/XGBoost/CatBoost; chemprop D-MPNN; TabPFN-2 and TabPFN-2.5 run in their designed regime (dense curated descriptors, full training set, no subsampling) — we retire an earlier internal comparison that handicapped TabPFN with 1,252 sparse features and out-of-memory subsampling; TabFM zero-shot. TabPFN-3 weights are license-token-gated and were not runnable in this environment; this is disclosed rather than substituted. Published external numbers (e.g. TransTg) are cited as reference points only, since their exact splits are not reproducible; no superiority claim is made against any number whose protocol we cannot replicate.

TABLE I
T1: mean MAE (K) by split family (identical frozen splits).

Model	Random MAE (K)	Class-out MAE
AFPE stack (ours, commercial-clean)	–	–
AFPE + TabPFN meta (ceiling; non-commercial)	21.10	30.36
Tuned GBM trio (LGBM+XGB+CatBoost)	23.25	33.48
TabPFN-2.5 zero-shot (non-commercial)	20.90	36.33
TabPFN-2 zero-shot (non-commercial)	22.64	30.55
Chemprop D-MPNN	25.28	42.44

TABLE II
T1: paired ΔMAE vs. ours (positive favors ours); † = does not survive FDR.

Family	Comparator	ΔMAE (K)	95% CI	Verd
butina	TabPFN-2 zero-shot (non-commercial)	-1.16	[-2.47, +0.04]	TIE

V. Results

A. Polymer glass transition (T1)

Table I reports mean MAE per split family under identical frozen splits; Table II reports paired verdicts against our commercially-clean stack with 95% cluster-bootstrap confidence intervals and pre-registered effect thresholds. Verdicts are three-valued (WIN/TIE/LOSS from the comparator's perspective against ours) and losses survive into print.

B. Multi-property engine (T2)

Table III reports five molecular-dynamics-computed properties. We pre-declared this regime ($n \approx 1$k) as favorable to tabular foundation models; results are reported symmetrically with replicate-noise floors, and verdicts carry FDR control.

C. Thermoset transfer (T3)

On 15 verified cured-network systems under nested leave-one-hardener-family-out evaluation, the incumbent aromatic-fraction baseline achieves 30.87 °C grouped MAE; a Gaussian-process model on fingerprints reaches 37.94 °C. We report this transfer honestly as weak: cured-network glass transition is under-determined by hardener structure alone (identical hardeners span $\sim$100 °C across epoxy/stoichiometry/cure contexts), and laboratory calibration is the designed path to a real thermoset model.

D. Rediscovery without cheating and throughput (T4)

Holding out five elite high-glass-transition hardeners from all training and ranking them among 50 property-matched, topologically-dissimilar decoys each (DUD-E style) with a physics-grounded rigidity score, the engine achieves AUROC 0.7252 with permutation p =0.008549572521373932 (median active percentile 10.2%; EF@5% = 3.923; BEDROC = 0.168). Against our pre-registered bar (top-5% median, EF@5%$\geq$5, BEDROC$\geq$0.5), this is a statistically significant enrichment signal that *does not* clear the full discovery threshold — reported as such. Screening throughput is 7,672 candidates/s on a single core over 200,000 molecules; fleet-scale figures are linear extrapolations and are labeled as such.

TABLE III
T2: RADONPY SURROGATE-OF-SIMULATION MAE WITH REPLICATE NOISE FLOORS.

T2 results: NOT RUN

VI. LIMITATIONS

(1) T2 targets are simulator outputs, not experiments; the engine is validated there as a surrogate of molecular dynamics. (2) The thermoset anchor set is small (n =15) and context-confounded; no state-of-the-art claim is made. (3) The rediscovery signal is significant but below our pre-registered discovery bar; with only five actives, early-enrichment metrics are high-variance. (4) No wet-laboratory performance data exist yet for any recommended formulation; all coating-performance quantities remain MODELED or ABSENT pending the laboratory program. (5) TabPFN-3 could not be included (license-gated weights); TabPFN-2.5 is the strongest foundation-model baseline run.

VII. ROADMAP

The laboratory year converts the evidence base: measured pot-life, VOC, adhesion, salt-fog, and cathodic disbondment flow back through a structured capture template, promoting MODELED claims to REAL, calibrating the thermoset model on in-context data, and closing an active-learning loop over the formulation space. The prediction engine extends to additional property heads under the same frozen-protocol discipline, and the commercially-clean stack remains the product path with foundation models tracked as licensed ceiling references.

VIII. CONCLUSION

For safety-critical formulation programs, the reliability of the *process* that produced a number matters more than the number. We have shown that a leakage-hardened, pre-registered, symmetric-reporting protocol can coexist with competitive performance, honest foundation-model comparisons, and a deterministic path from customer envelope to laboratory-executable formulation. The discipline itself is the contribution.

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